

Skill Swap

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Group Number: 02

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CERTIFICATE

This is to certify that the project report entitled “**Skill Swap**” has been carried out by Ms. Shruti Hansaliya(202206100110060), Ms. Divya Ghorl (202206100110061) and Mr. Krish Patel (202206100110107) under my guidance in partial fulfillment of the requirement for the **Master of Science in Information Technology** during the academic year 2025-26.

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**BABU MADHAV INSTITUTE OF
INFORMATION TECHNOLOGY**

Table of Contents

Chapters	Particulars		Page no.
1	Introduction		
	1.1	Problem Definition	6
	1.2	Purpose	6
	1.3	Scope	6
2	Overall Description		
	2.1	Product Perspective/ Environment Description	
	2.1.1	Hardware Interface / Hardware Specification	7
	2.1.2	Software Interface / Software Specification	7
3	System Specific Requirements		
	3.1	Functional Requirement	8-12
	3.2	Non-functional Requirement	13
4	System Analysis		
	4.1	Use case Diagram	14
	4.2	Activity Diagrams	15-19
5	System Design		
	5.1	Database Design	20-21
	5.1.2	Data Dictionary	22-36
6	System Implementation		
	6.1	Screenshots	37-51
7	Testing		
	7.1	Test Cases	52-53
8	Future Enhancement		54
9	Conclusion		55
10	Bibliography		56

List of Figures and Diagrams

Figure No.	Figure Description	Page No.
1	USE CASE DIAGRAM	14
2	ACTIVITY DIAGRAM FOR ADMIN MANAGE CATEGORY, SKILL, CITY, SUBSCRIPTIONS	15
3	ACTIVITY DIAGRAM FOR MANAGE UNAUTHORIZED MEMBER	16
4	ACTIVITY DIAGRAM FOR USER REQUEST AND SKILL SWAP	17
5	ACTIVITY DIAGRAM FOR COMMUNICATION FEATURES FOR SESSION	18
6	ACTIVITY DIAGRAM FOR FEEDBACK	18
7	ACTIVITY DIAGRAM FOR PURCHASE SUBSCRIPTIONS	19

List of Tables

Table No.	Table Description	Page No.
1	City	22
2	SkillCategory	22
3	Skill	23
4	SubscriptionPlan	23
5	User	24
6	UserSkill	25
7	Subscription	26
8	Payment	27
9	Request	28
10	SkillSwap	29
11	Calls	30
12	Chat	31
13	Materials	31
14	Video	32
15	Feedback	33
16	Notification	34
17	Report	35
18	SkillRequest	36

Chapter 1: Introduction

1.1 Problem Definition

Many people have skills they can teach but lack the money or resources to learn new ones. Existing platforms are costly and one-way (teacher to student). There's a need for a free, peer-to-peer system where users can exchange skills directly. The Skill Swap React app solves this by allowing users to list skills they can teach and skills they want to learn, and then match with others for mutual exchange. The system connects skill seekers and providers, allows secure account and skill management, provides admin oversight, facilitates interaction and learning sessions.

1.2 Project Purpose

The purpose of this project is to develop an online **Skill Swap Platform** that allows users to share and exchange skills with other members in a secure and structured way. The system enables members to register, verify their skills, search for other skilled members, and request skill swap sessions. It also provides communication tools such as chat and video calls to support learning interactions. Admin controls proper management of subscriptions, payments, and reported users. Overall, the platform promotes skill development, collaboration, and community learning.

1.3 Project Scope

The scope of this project is limited to **users within a single state**.

Chapter 2: Overall Description

2.1 Product Perspective/Environment Description

The Skill Swap System is a web-based platform that allows people within one state to exchange skills in a safe and organized way. It works through a web browser and supports two types of users: Admin and Members. Members can register, create profiles, verify their skills, search for other members, request skill swaps, chat, make video calls, share learning materials, and give feedback after sessions. Admin manages users, skills, subscriptions, payments, and reported accounts. The system uses email and OTP for secure login, online payment for subscriptions, and stores all data in a central database to ensure smooth and secure operation.

2.1.1 Hardware Interface/ Hardware Specification

Processor: i5
Clock speed: 5200MHz
RAM: 16GB

2.1.2 Software Interface/ Software Specification

Technology used:
Frontend: HTML, CSS, React.js(Frontend framework)
Backend: Node.js, Express.js
Database: MongoDB
Scripting language: JavaScript
Development tools:
Visual Studio Code (VS Code) for coding and development
MongoDB Compass/Atlas for database management
Postman for API testing
Google Chrome/Firefox for application testing and debugging

Chapter 3: System Specific Requirements

3.1 Functional Requirements

User Account Management

Requirement No.	Description	Comment
FR1	Members can register by giving their skills, email, and profile details. Registration is completed through OTP and the system sends a welcome/confirmation notification using mail.	Registration page
FR2	Members (Admin & normal members) can log in via email and password.	Log in page
FR3	If a member forgets the password, then OTP is sent through the mail for the Member's verification and Member can generate new password.	Forget Password Page
FR4	Member can also change the password.	Change password page
FR5	After login Admin and Members can update their personal details.	Profile page

Admin Control

Requirement No.	Description	Comment
FR1	Members can report fake accounts and Notification goes to admin then admin blocks/reviews the user.	Report User Page in member dashboard
FR2	Admin can block unauthorized members after rechecking their profile.	Complaint page in admin dashboard
FR3	All subscription payments made by Members are collected and stored in the admin's account that can be view by admin.	View payment page
FR4	Admin can manage (add, update and delete) skill categories, skills and subscription plan.	Manage Skill category page, Manage Skill page, Manage Subscription page

Skill Verification

Requirement No.	Description	Comment
FR1	Members can upload certificates or documents for skills they claim.	Profile page
FR2	Admin view member's detail.	Member page

Search & Filter

Requirement No.	Description	Comment
FR1	Members can search for other Members by skill or by rating.	Skill Swap page

Member Request

Requirement No.	Description	Comment
FR1	A Member can send a request to connect with another Member.	Skill Swap page
FR2	The recipient can accept or reject or cancel the request. Notifications are sent to both members.	Show request page
FR3	Both Members must confirm before starting a skill swap session.	Confirmation page

Communication Features

Requirement No.	Description	Comment
FR1	Participants who accepted requests can chat with each other.	Chat page
FR2	Participants can start a video call session after acceptance.	Video call page
FR3	Participants can upload materials/video but share only with the session partner (not publicly).	Upload material page

Feedback, Rating & Report(complaint)

Requirement No.	Description	Comment
FR1	After a session, participants can provide feedback and rate each other.	Feedback page
FR2	Participants can report inappropriate behavior, which goes to admin for review.	Report user page

Subscription

Requirement No.	Description	Comment
FR1	Members get 2 free skill swaps.	Skill Swap Packs page
FR2	Once the free swaps are used, the Member must purchase additional swaps (2 swap pack, 3 swap pack).	Swap Subscription page
FR3	Pricing based on swaps in subscription and payment is done through online.	Swap Pricing page

History

Requirement No.	Description	Comment
FR1	Members can view their past skill swaps, including date, partner, and skill details.	History page
FR2	Members can view all given and received feedback & ratings.	Feedback history page
FR3	Members can see how many free swaps they used and what swap packs they bought.	Subscription history page

New Skill Request Management

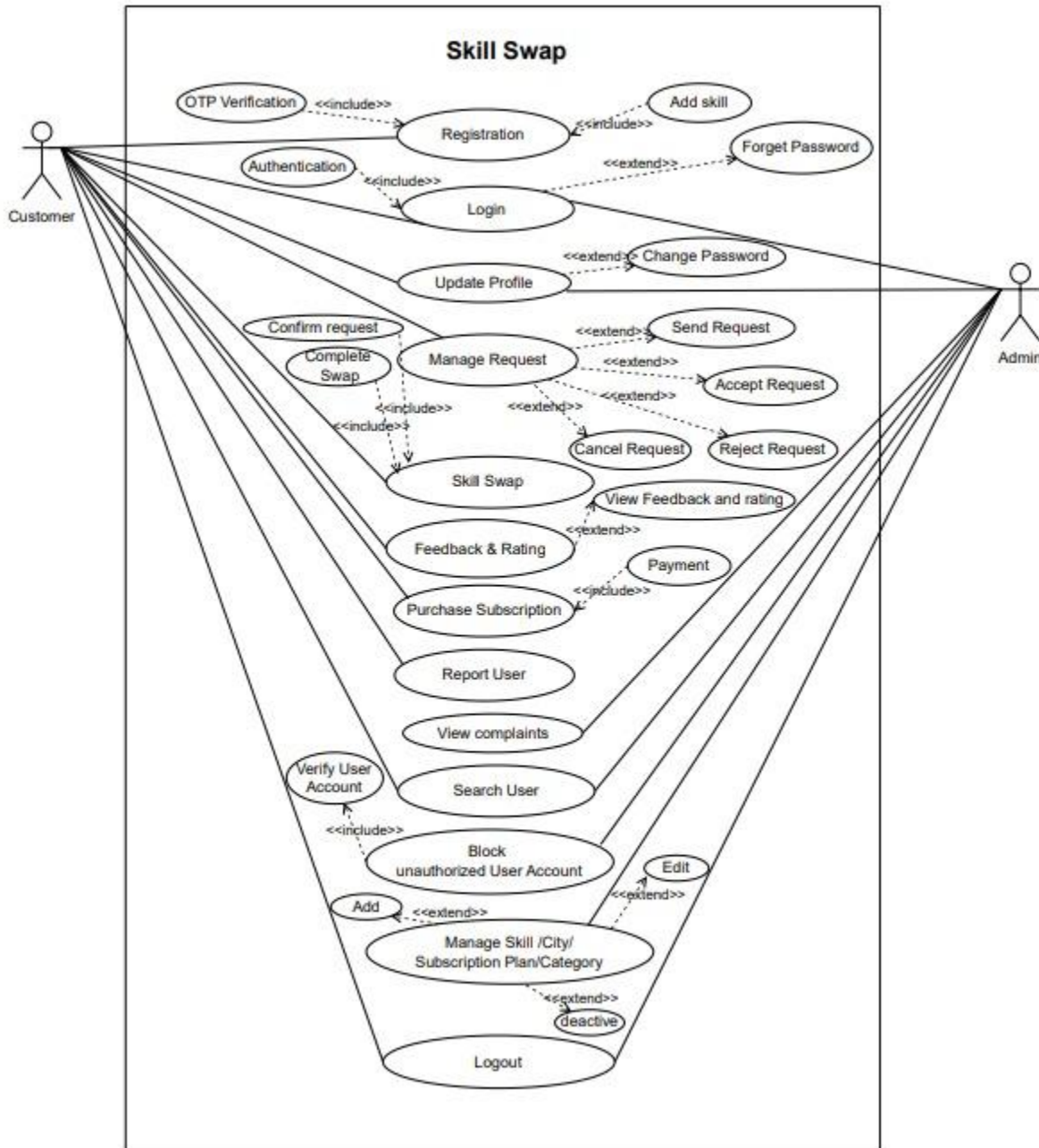
Requirement No.	Description	Comment
FR1	Members can send a request to the admin to add a new skill under a specific category.	Add Skill page
FR2	Admin can view the skill requests sent by members and either approve or reject them. • If approved, Admin adds the new skill under the selected category and notifies the member. • If rejected, Admin sends a proper reason to the member for rejecting the skill request	Skill Request from User page

3.2 Non- Functional Requirements

Requirement No.	Description	Comment
NFR1	Passwords encrypted (bcrypt/argon2). If User can login with wrong username or password after three wrong attempt it will lock user to login into system for few second.	Security
NFR2	The system should be responsive across mobile, tablet, and desktop devices.	Responsiveness

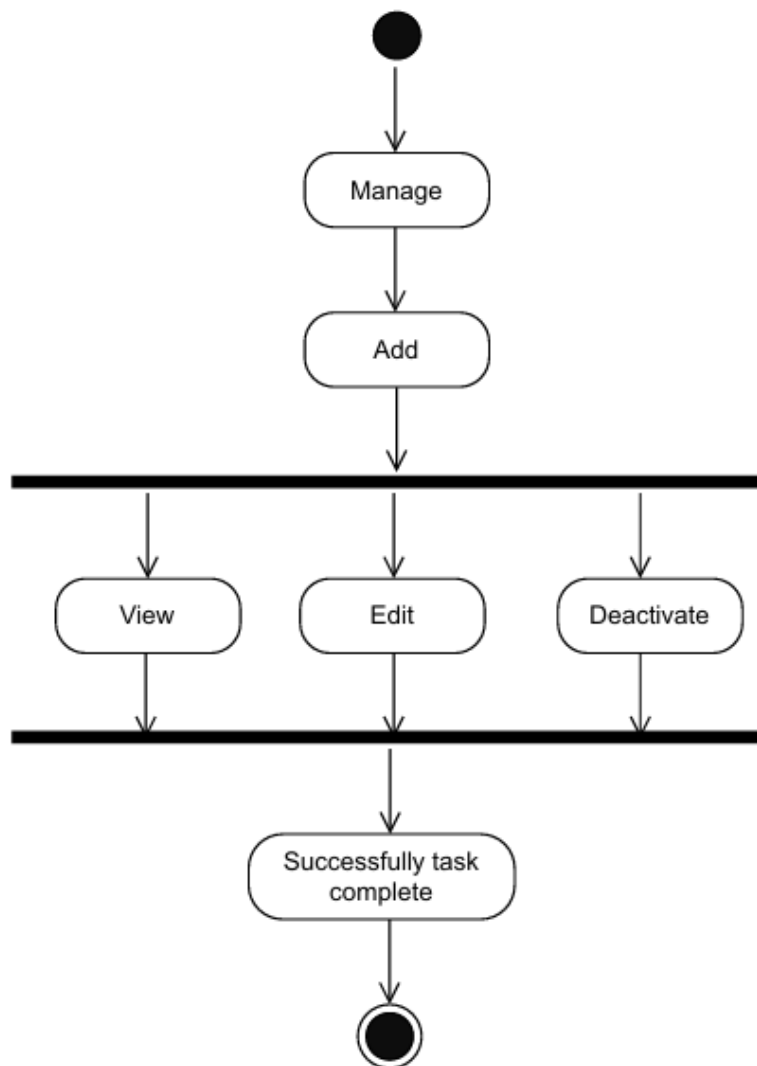
Chapter 4: System Analysis

4.1 Use Case Diagrams

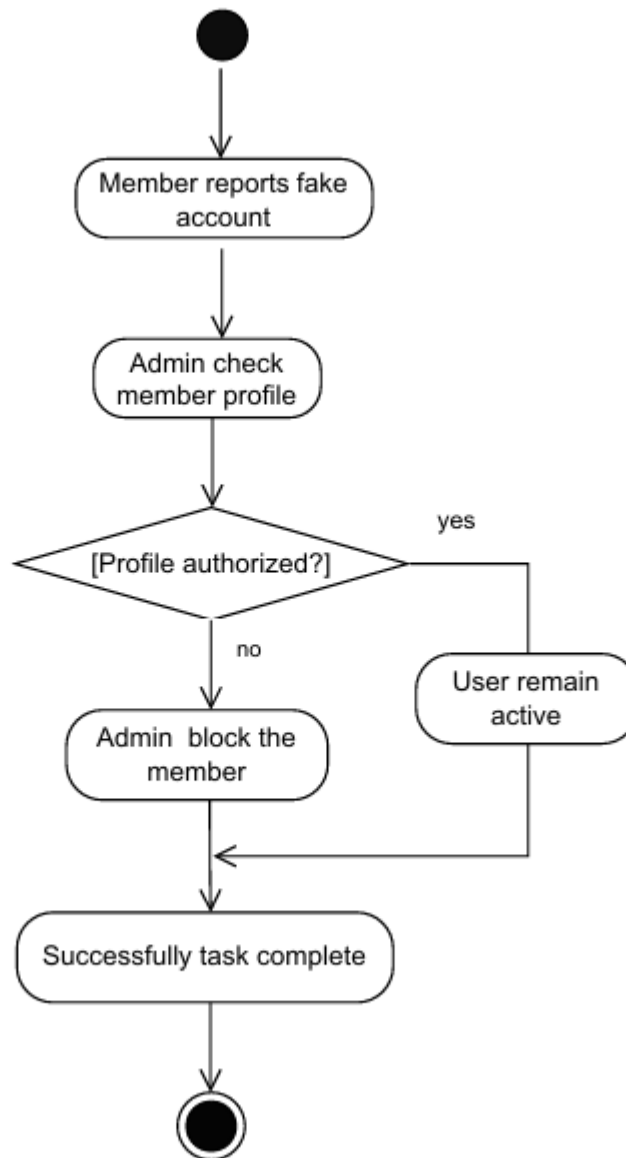


4.2 Activity Diagrams

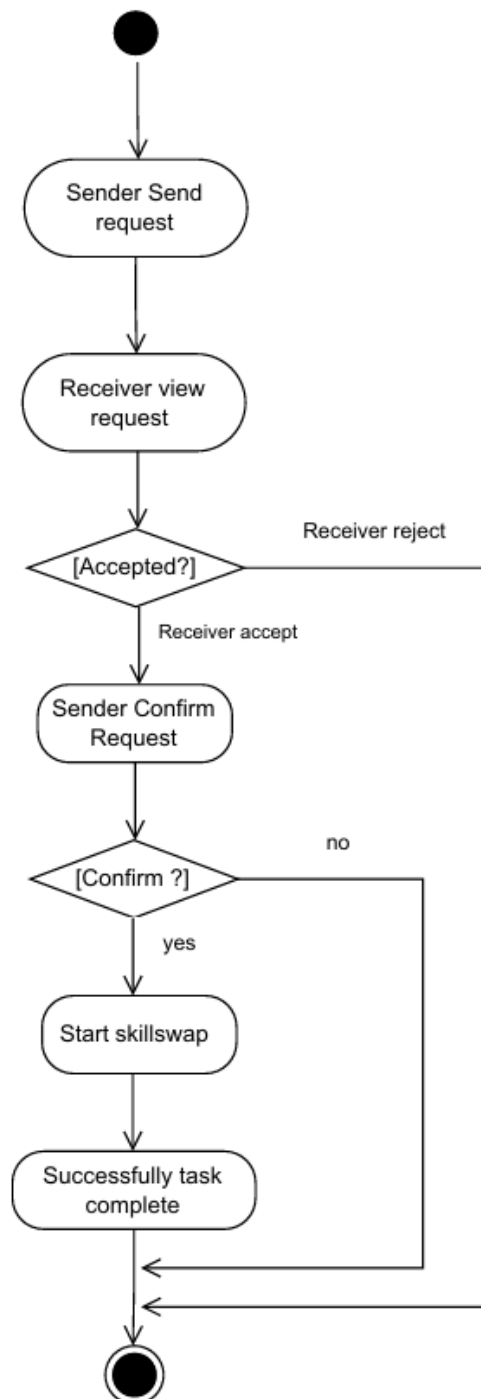
4.2.1 Admin manage Category/Skill/City/Subscriptions



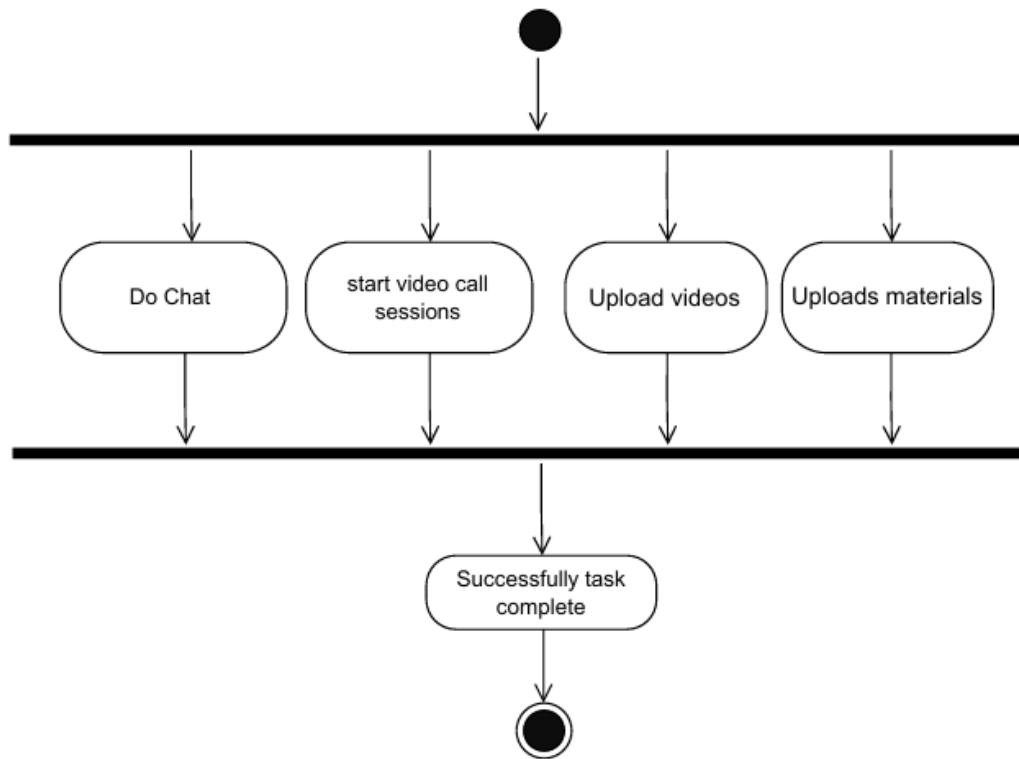
4.2.2 Admin manage unauthorized member



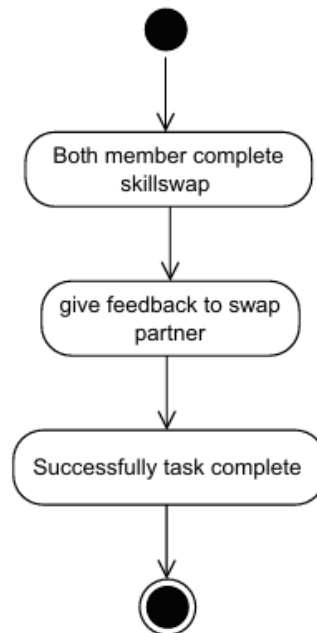
4.2.3 User Request and skill swap



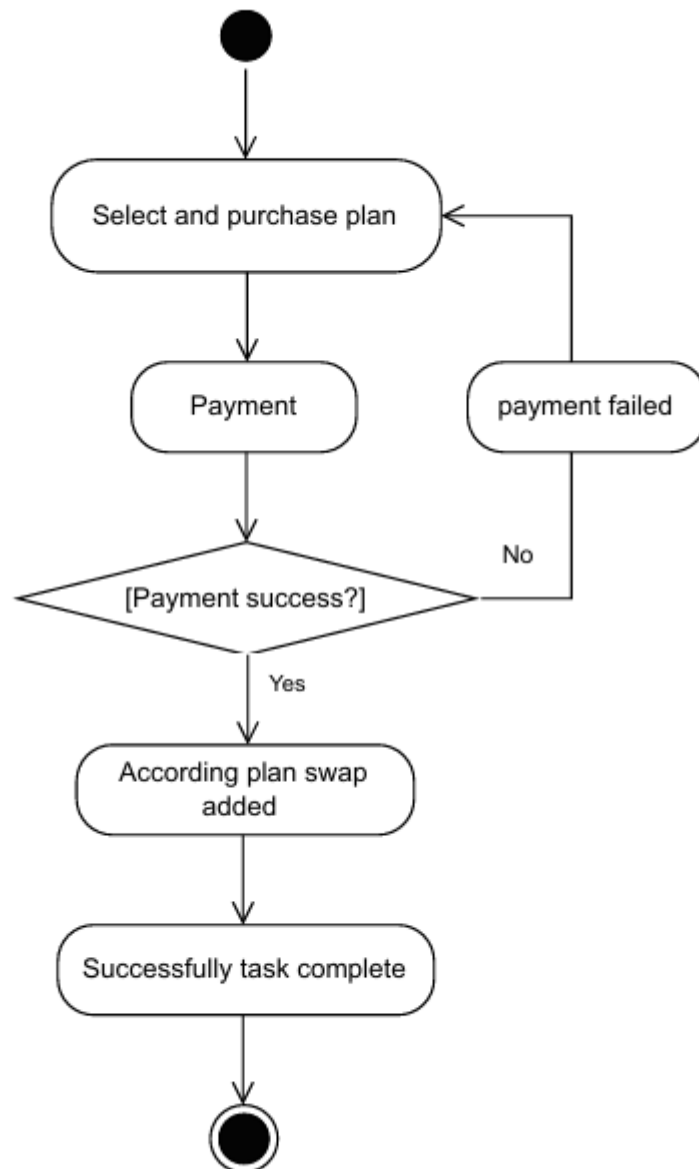
4.2.4 Communication Features for session



4.2.5 Feedback



4.2.6 Purchase Subscriptions



Chapter 5: System Design

5.1 System Design

City (_id [ObjectId], CityName, Status)

SkillCategory (_id [ObjectId], CategoryName, Status, Image)

Skill (_id [ObjectId], CategoryId [FK], Name, Status)

SubscriptionPlan (_id [ObjectId], Name, Status, SwapLimit, Price, CreatedAt, UpdatedAt)

User (_id [ObjectId], Username, Email, Password, DateOfBirth, Gender, StreetNo, Area, City [FK], ContactNo, Bio, RegisteredDate, Status, IsVerified, Role, ProfileImageUrl)

UserSkill (_id [ObjectId], UserId [FK], SkillId [FK], CertificateURL, ContentFileURL, Source, SkillAvailability, AddedDate, CertificateStatus)

Subscription (_id [ObjectId], UserId [FK], PlanId [FK], IsFreePlan, SwapsRemaining, StartDate, ExpiryDate, Status, PaymentStatus, RazorpayOrderId, RazorpayPaymentId, RazorpaySignature)

Payment (_id [ObjectId], UserId [FK], PlanId [FK], RazorpayOrderId, RazorpayPaymentId, RazorpaySignature, Amount, Currency, Status, CreatedAt, UpdatedAt)

Request (_id [ObjectId], SenderId [FK], ReceiverId [FK], SkillToLearnId [FK], SkillToTeachId [FK], Status, CreatedAt, UpdatedAt)

SkillSwap (_id [ObjectId], RequestId [FK], Status, SenderConfirmed, ReceiverConfirmed, CreatedAt, CompletedAt)

Call (_id [ObjectId], SwapId [FK], CallerId [FK], ReceiverId [FK], Status, StartedAt, EndedAt, DurationSeconds, CreatedAt, UpdatedAt)

Chat (_id [ObjectId], SwapId [FK], SenderId [FK], ReceiverId [FK], Message, SentAt, Status)

Material (_id [ObjectId], SwapId [FK], UserId [FK], FileURL, UploadedAt)

Video (_id [ObjectId], SwapId [FK], UserId [FK], FileURL, OriginalName, Description, CreatedAt)

Feedback (_id [ObjectId], SwapId [FK], SenderId [FK], ReceiverId [FK], Rating, Comments, Date, CreatedAt, UpdatedAt)

Notification (_id [ObjectId], UserId [FK], Message, Type, Link, IsRead, CreatedAt)

Report (_id [ObjectId], Reporter [FK], ReportedUser [FK], Reason, Description, Evidence, Status, AdminNote, ActionTaken, SuspendUntil, CreatedAt, UpdatedAt)

SkillRequest (_id [ObjectId], UserId [FK], Username, SkillName, Message, Status, AdminReply, ReadByAdmin, CreatedAt, UpdatedAt)

5.1.1 Data Dictionary

City

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for city
2	cityName	String	—	Unique, Not Null	Name of the city
3	status	String	—	Enum (Active, Inactive)Default Active	City status

SkillCategory

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	—	PK	Unique Id for category ID
2	CategoryName	String	100	Not Null	Name of skill category(music)
3	status	String	—	Enum (Active, Inactive)Default Active	Category status
4	image	String	—	Nullable	Category image path

Skill

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	—	PK	Unique Id for skill id
2	CategoryId	Number	—	FK, Not Null	Reference to SkillCategory
3	Name	String	150	Not Null	Name of skill (React.js)
4	Status	String	—	Enum (Active, Inactive) Default Active	Skill status

SubscriptionPlan

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	—	PK	Unique Id for plan ID
2	Name	String	—	Not Null	Plan name
3	status	String	—	Enum (Active, Inactive) Default Active	Plan status
4	SwapLimit	Number	—	Default 0	Allowed swaps
5	Price	Decimal	—	Default 0.0	Plan price
6	createdAt	DateTime	—	Default Now()	Record creation
7	updatedAt	DateTime	—	Default Now()	Record update

User

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	—	PK	Unique Id for user ID
2	Username	String	100	Not Null, Unique	User name
3	Email	String	255	Unique	User email
4	Password	String	—	Encrypted	User password
5	DateOfBirth	Date	—	Not Null	User DOB
6	Gender	String	—	Enum (Male, Female, Other), Not Null	User gender
7	StreetNo	String	—	Not Null	Street address
8	Area	String	—	Not Null	Area name
9	City	ObjectId	24	FK, Not Null	City reference
10	ContactNo	String	10	Unique	Phone number
11	Bio	String	-	Null	User's short bio
12	RegisteredDate	DateTime	—	Default Now ()	Registration date
13	Status	String	—	Enum (Active, Inactive), Default Active	User status
14	IsVerified	Boolean	—	Default false	OTP verification
15	Role	String	—	Enum (member, Admin), Default member	User role
16	ProfileImageURL	String	—	Nullable	Profile image

UserSkill

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for user skill ID
2	UserId	ObjectId	24	FK, Not Null	User reference
3	SkillId	Number	—	FK, Not Null	Skill reference
4	CertificateURL	String	—	Nullable	Certificate file
5	ContentFileURL	String	—	Nullable	Learning content
6	Source	String	150	Nullable	Skill source
7	SkillAvailability	String	—	Enum (Available, Unavailable), Default Available	Skill availability
8	AddedDate	Date	—	Default Now()	Added date
9	CertificateStatus	String	—	Enum (Pending, Verified, Rejected), Default Pending	Verification status

Subscription

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique subscription ID
2	UserId	ObjectId	24	FK, Not Null	User reference
3	PlanId	ObjectId	24	FK, Null	Subscription plan reference
4	IsFreePlan	Boolean	—	Default false	Free plan flag
5	SwapsRemaining	Number	—	Not Null	Remaining swaps
6	StartDate	Date	—	Default Now()	Subscription start
7	ExpiryDate	Date	—	Nullable	Subscription expiry
8	Status	String	—	Enum (Active, Upcoming, Expired),Default Active	Subscription status
9	PaymentStatus	String	—	Enum (Pending, Success, Failed),Default Pending	Payment status
10	RazorpayOrderId	String	—	Nullable	Payment order ID
11	RazorpayPaymentId	String	—	Nullable	Payment transaction ID
12	RazorpaySignature	String	—	Nullable	Payment signature

Payment

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for payment
2	userId	ObjectId	24	FK, Not Null	Reference to User
3	planId	ObjectId	24	FK, Not Null	Reference to SubscriptionPlan
4	razorpay_order_id	String	—	Nullable	Razorpay order id
5	razorpay_payment_id	String	—	Nullable	Razorpay payment id
6	razorpay_signature	String	—	Nullable	Razorpay signature
7	amount	Number	—	—	Payment amount
8	currency	String	—	Default INR	Currency type
9	status	String	—	Enum(Created ,Paid , Failed),Default Created	Payment status
10	createdAt	DateTime	—	Default Now()	Record creation time
11	updatedAt	DateTime	—	Default Now()	Record update time

Request

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for request
2	SenderId	ObjectId	24	FK, Not Null	(Request sender) User reference
3	ReceiverId	ObjectId	24	FK, Not Null	(Request receiver) User reference
4	SkillToLearnId	ObjectId	24	FK, Not Null	(Skill user wants to learn) UserSkill reference
5	SkillToTeachId	ObjectId	24	FK, Nullable	(Skill user offers to teach) UserSkill reference
6	Status	String	15	Enum (Pending, Accepted, Rejected), Default Pending	Pending / Accepted / Rejected
7	createdAt	DateTime	—	Default Now()	Request creation time
8	updatedAt	DateTime	—	Default Now()	Request update time

SkillSwap

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique skill swap ID
2	RequestId	ObjectId	24	FK, Not Null	(Related request ID)Request reference
3	Status	String	—	Enum (Active, Completed, Cancelled),Default Active	Swap status
4	SenderConfirmed	Boolean	—	Default false	Sender confirmation
5	ReceiverConfirmed	Boolean	—	Default false	Receiver confirmation
6	CreatedAt	Date	—	Default Now()	Swap start time
7	CompletedAt	Date	—	Nullable	Swap completion time

Calls

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for call
2	swapId	ObjectId	24	FK, Not Null	Reference to SkillSwap
3	callerId	ObjectId	24	FK, Not Null	Reference to (caller) User
4	receiverId	ObjectId	24	FK, Not Null	Reference to (receiver) User
5	status	String	20	Enum(ongoing, completed, missed, cancelled) ,Default ongoing	Call status
6	startedAt	Date	—	Default Now()	Call start time
7	endedAt	Date	—	Nullable	Call end time
8	durationSeconds	Number	—	Default 0	Call duration in seconds
9	createdAt	Date	—	Default Now()	Record creation time
10	updatedAt	Date	—	Default Now()	Record update time

Chat

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id
2	SwapId	ObjectId	24	FK, Not Null	Reference to SkillSwap
3	SenderId	ObjectId	24	FK, Not Null	Reference to (sender) User
4	ReceiverId	ObjectId	24	FK, Not Null	Reference to (receiver) User
5	Message	String	—	Not Null	Chat message
6	SentAt	Date	—	Default Now()	Message sent time
7	Status	String	15	Enum(Sent , Delivered , Read) ,Default Sent	Chat Status

Materials

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id
2	SwapId	ObjectId	24	FK, Not Null	Reference to SkillSwap
3	UserId	ObjectId	24	FK, Not Null	Reference to user
4	FileURL	String	—	Not Null	Uploaded material URL
5	UploadedAt	Date	—	Default Now()	Upload timestamp

Video

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique video ID
2	swapId	ObjectId	24	FK, Not Null	Skill swap reference
3	userId	ObjectId	24	FK, Not Null	User reference
4	fileUrl	String	—	Not Null	Video file URL
5	originalName	String	—	Nullable	Original file name
6	description	String	—	Nullable	Video description
7	createdAt	Date	—	Default Now()	Upload time

Feedback

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for feedback
2	SwapId	ObjectId	24	FK, Not Null	Reference to SkillSwap
3	SenderId	ObjectId	24	FK, Not Null	(Feedback given by user) User reference
4	ReceiverId	ObjectId	24	FK, Not Null	(Feedback received by user) User reference
5	Rating	Number	1	Not Null	Rating (1 to 5)
6	Comments	String	—	Nullable	Feedback comments
7	Date	Date	—	Default Now()	Feedback date
8	createdAt	Date	—	Default Now()	Record creation time
9	updatedAt	Date	—	Default Now()	Record update time

Notification

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for notification
2	userId	ObjectId	24	FK, Not Null	Reference to User
3	message	String	—	Not Null	Notification message text
4	type	String	—	Enum , Default general	Type of notification
5	link	String	—	Nullable	Redirection link
6	isRead	Boolean	—	Default false	Read status
7	createdAt	Date	—	Default Now()	Notification creation time

Report

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique Id for report
2	reporter	ObjectId	24	FK, Not Null	(User who reported) User reference
3	reportedUser	ObjectId	24	FK, Not Null	(User being reported) User reference
4	reason	String	—	Not Null	Reason for report
5	description	String	—	Nullable	Detailed explanation
6	evidence	Array	—	Nullable	Proof files (URL, mimeType)
7	status	String	—	Enum(pending ,in_review ,resolved ,rejected ,suspended), Default pending	Report Status
8	adminNote	String	—	Nullable	Admin remarks
9	actionTaken	String	—	Enum(none , warned , suspended , deleted), Default none	Action taken enum
10	suspendUntil	Date	—	Nullable	Suspension expiry date
11	createdAt	Date	—	Default Now()	Record creation time
12	updatedAt	Date	—	Default Now()	Record update time

SkillRequest

Sr. No	Field Name	Datatype	Size	Constraint	Description
1	_id	ObjectId	24	PK	Unique skill request ID
2	UserId	ObjectId	24	FK, Not Null	(User who requested new skill) User reference
3	Username	String	—	Not Null	Username of requester
4	SkillName	String	200	Not Null	Requested skill name
5	Message	String	2000	Nullable	Request message
6	Status	String	—	Enum (Pending, Approved, Rejected), Default Pending	Request status
7	AdminReply	String	2000	Nullable	Admin response
8	ReadByAdmin	Boolean	—	Default false	Admin read status
9	CreatedAt	DateTime	—	Default Now()	Request created time
10	UpdatedAt	DateTime	—	Default Now()	Request update time

Chapter 6: System Implementation

6.1 Screenshots

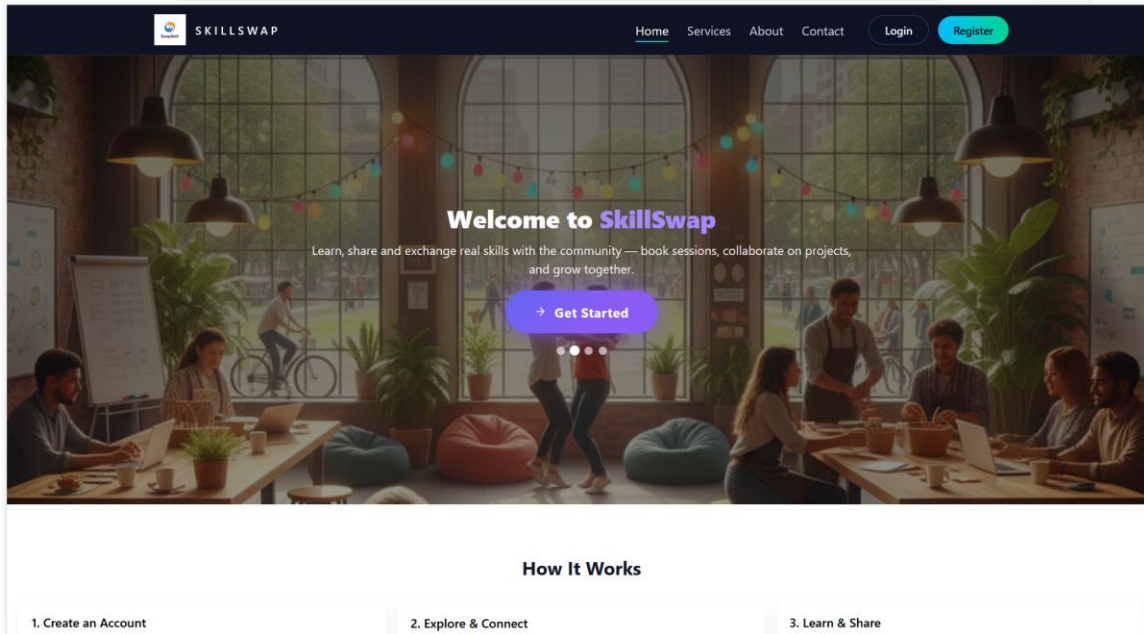


Figure 6.1.1: Home page

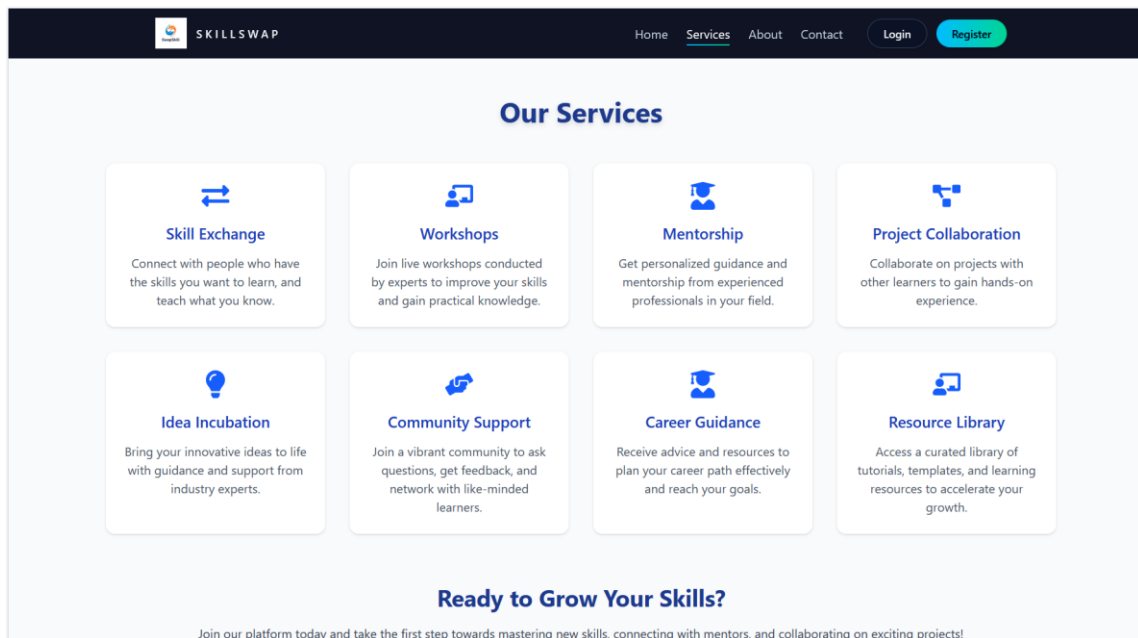


Figure 6.1.2: Our service page

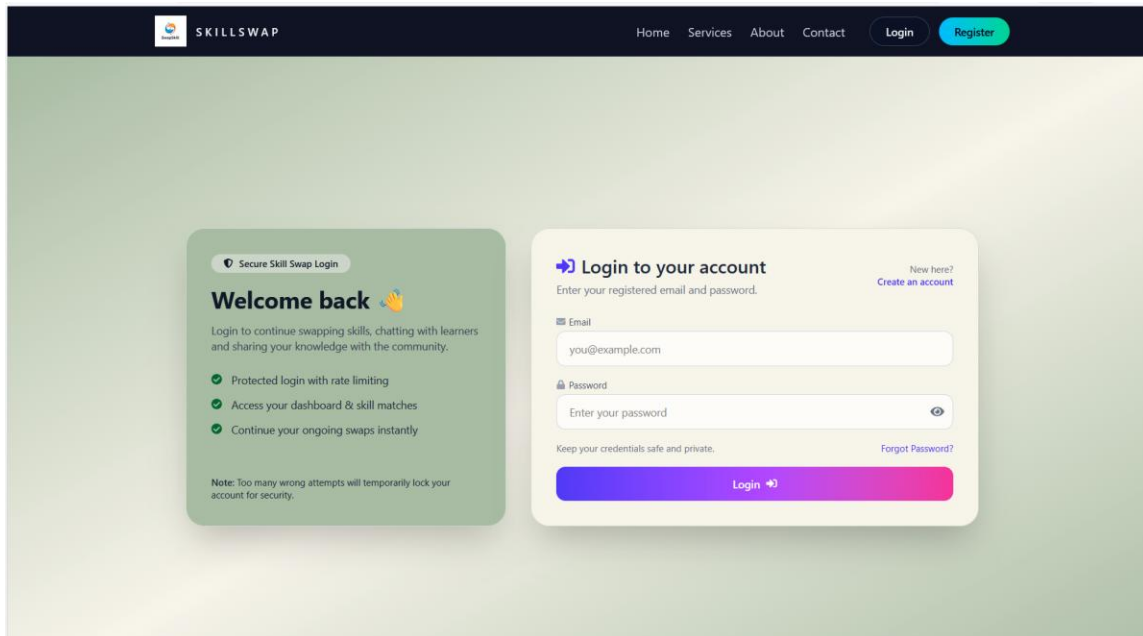


Figure 6.1.3: Login page

Figure 6.1.4: Registration page

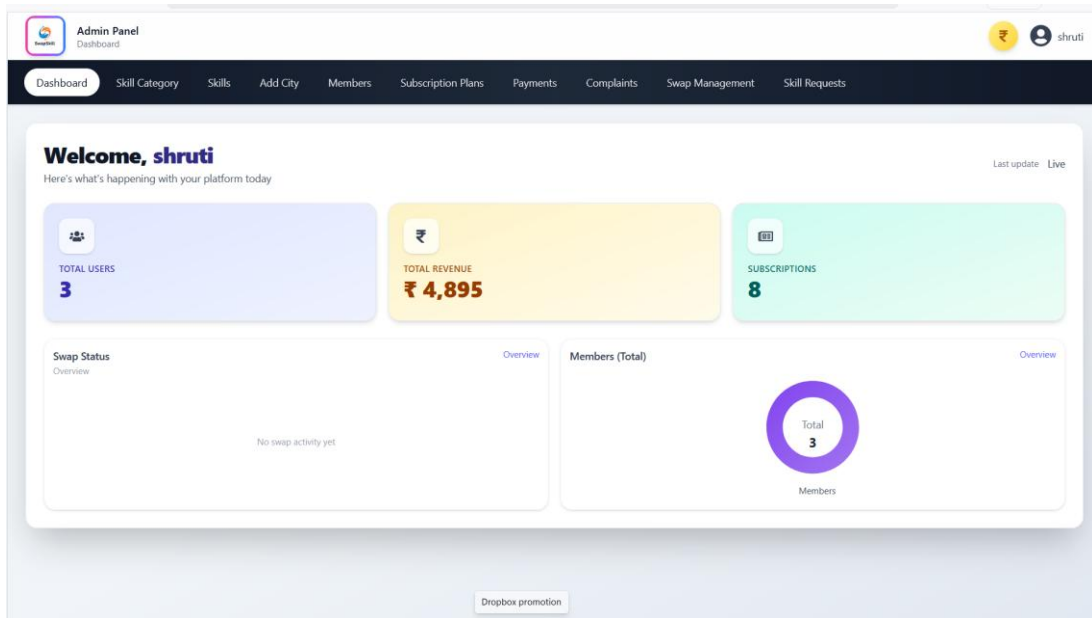


Figure 6.1.5: Admin Dashboard Overview page

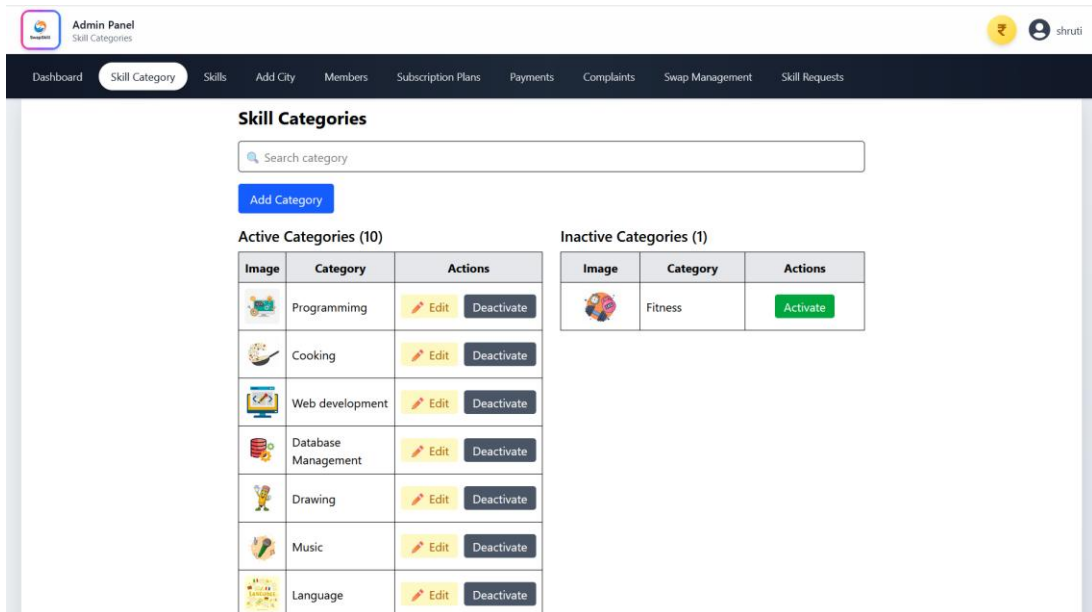


Figure 6.1.6: Add category page

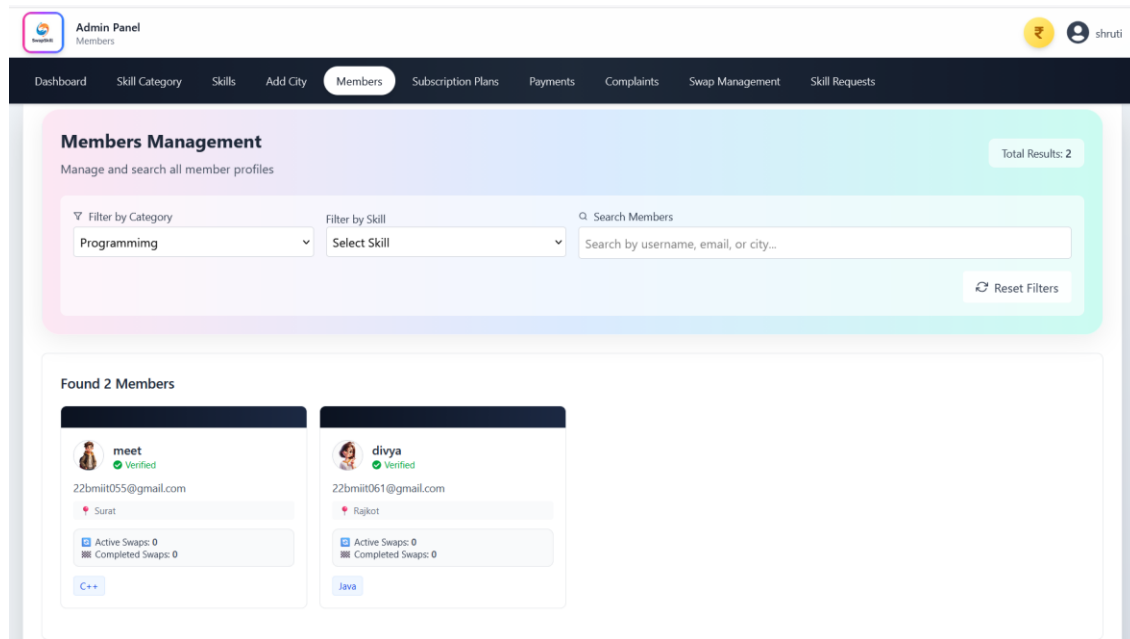


Figure 6.1.7: View Member page

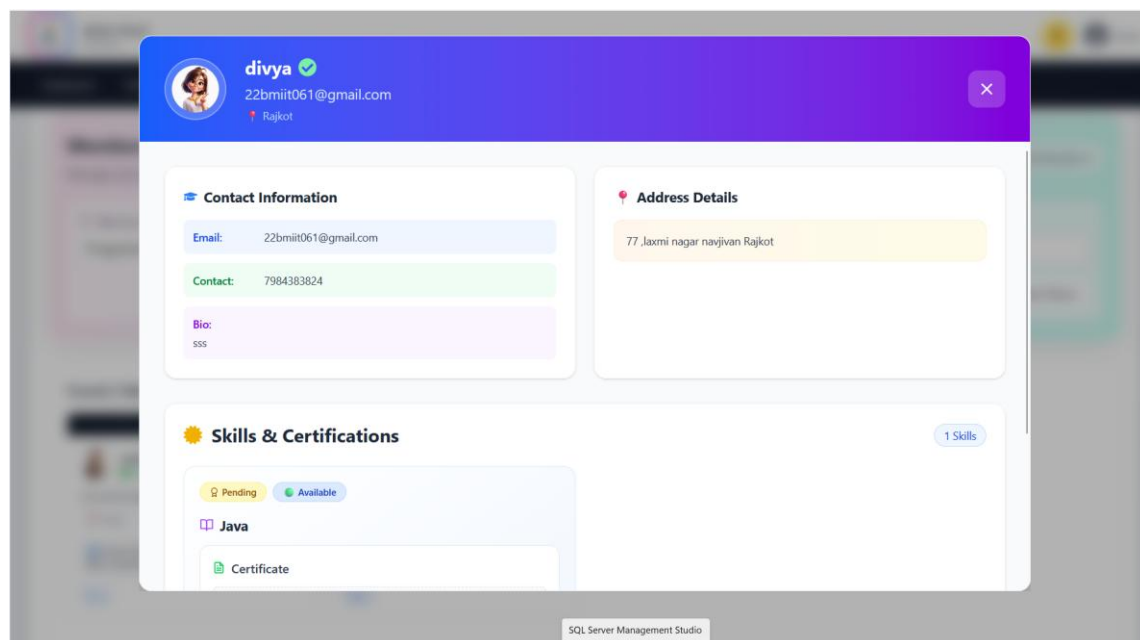


Figure 6.1.8: Member detail popup page

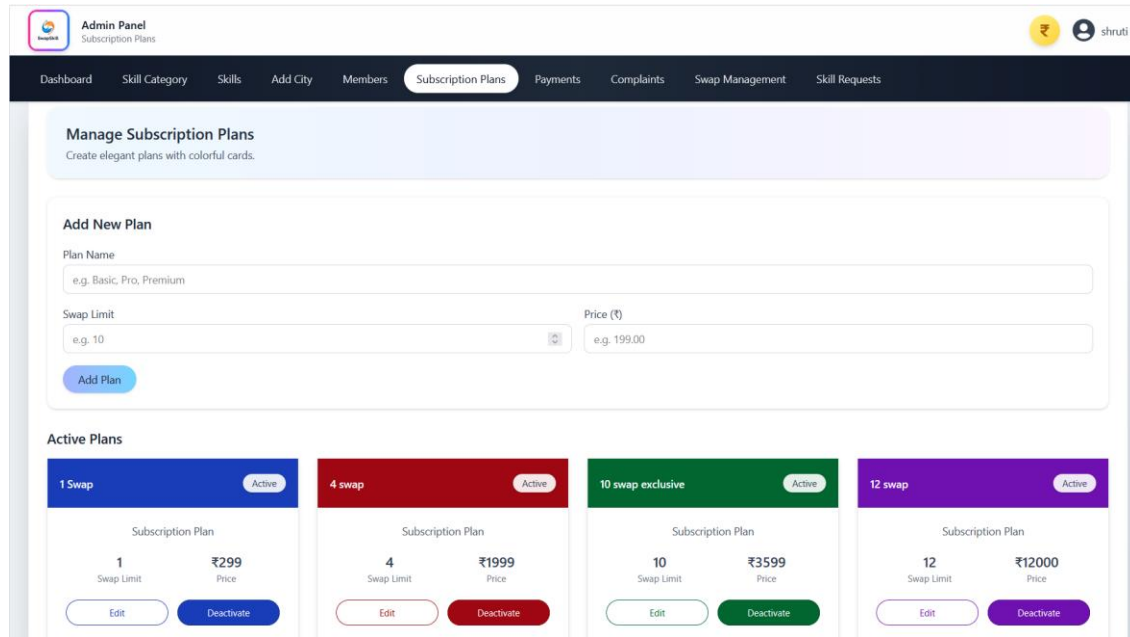


Figure 6.1.9: Subscription page

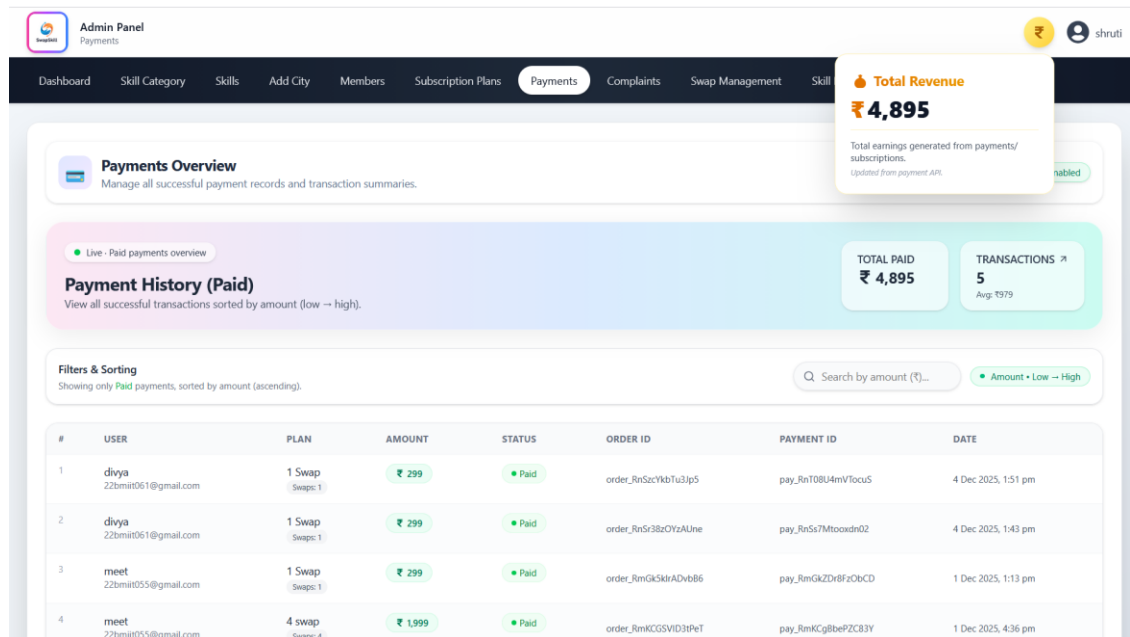


Figure 6.1.10: Payment Detail page

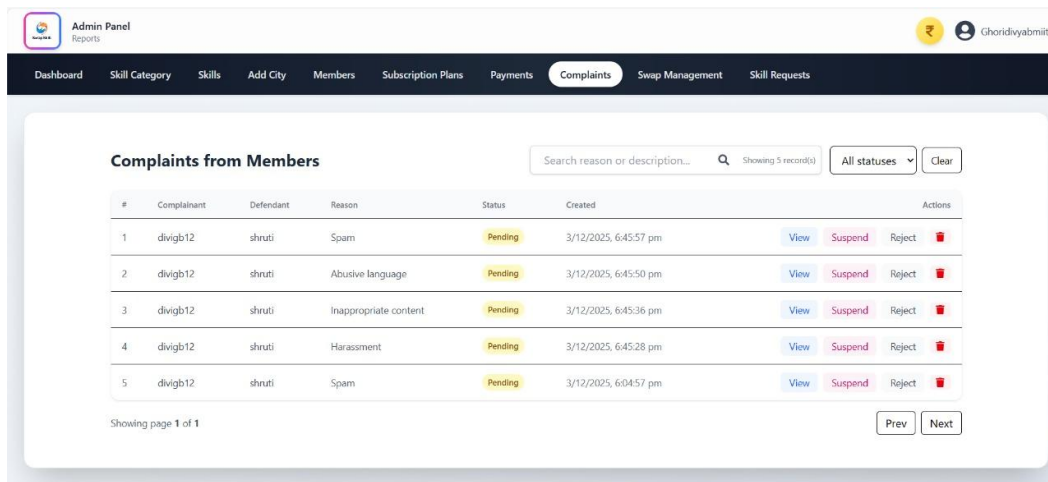


Figure 6.1.11: Manage complaint page

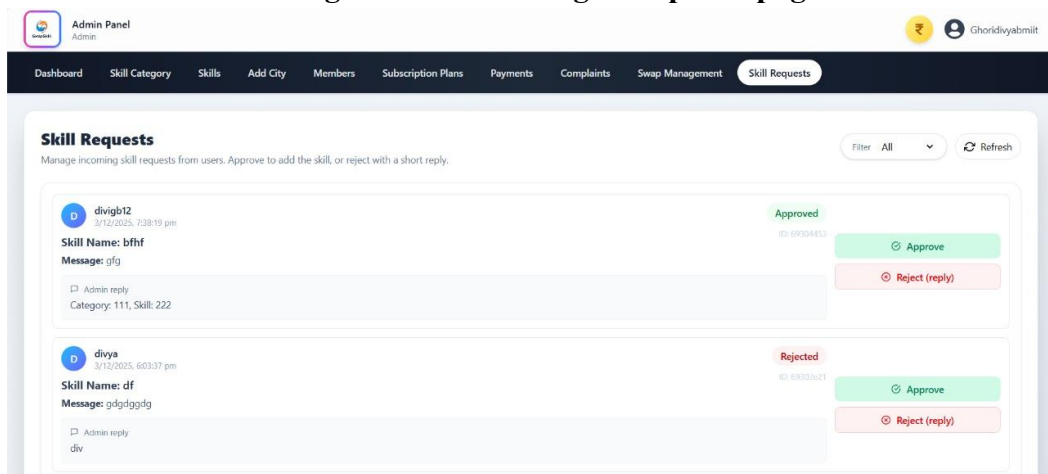


Figure 6.1.12: Manage Skill request page

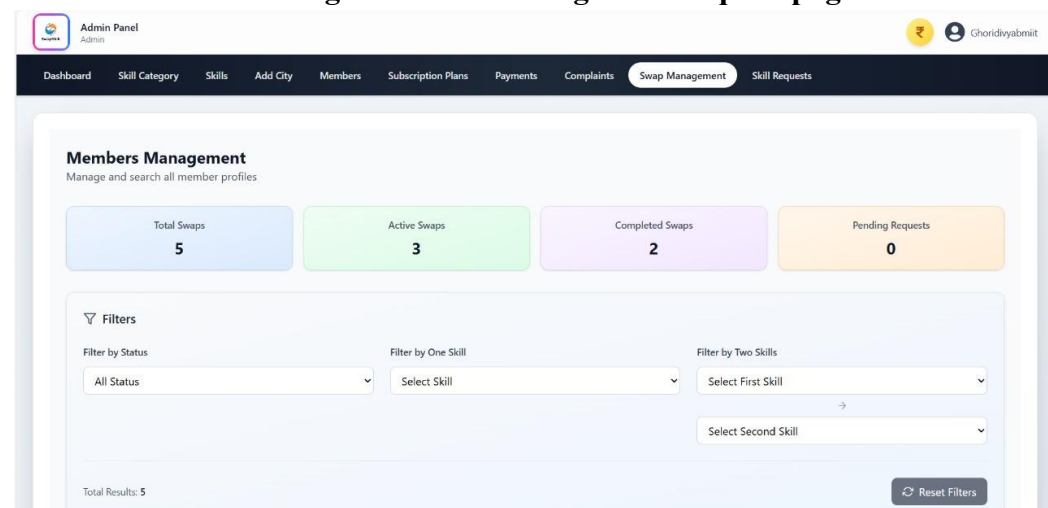


Figure 6.1.13: View SkillSwap page

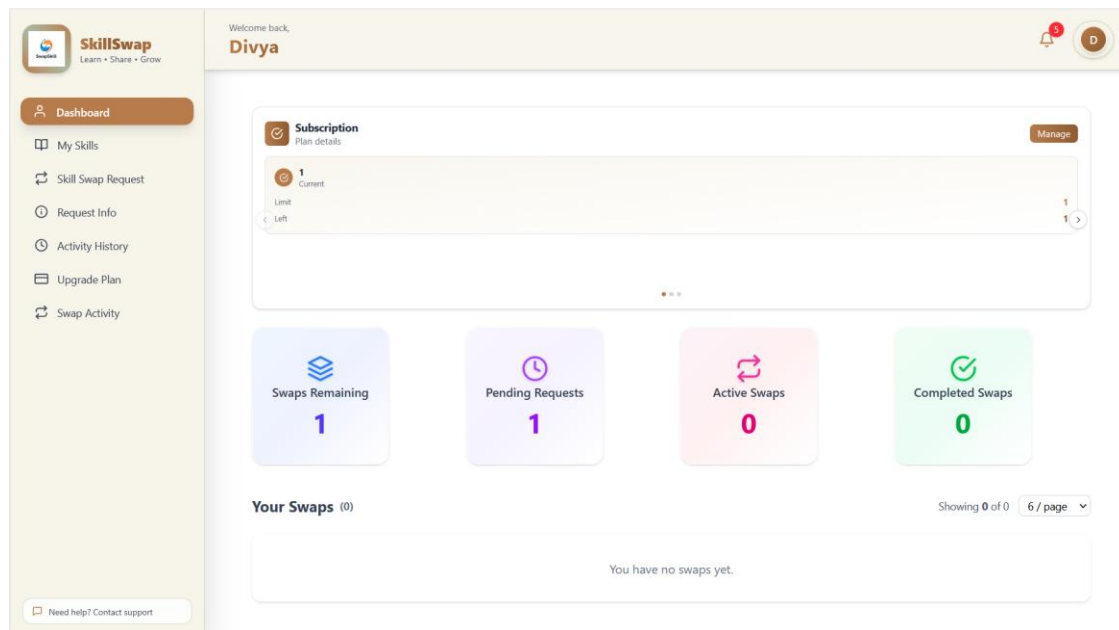


Figure 6.1.14: Dashboard page of member

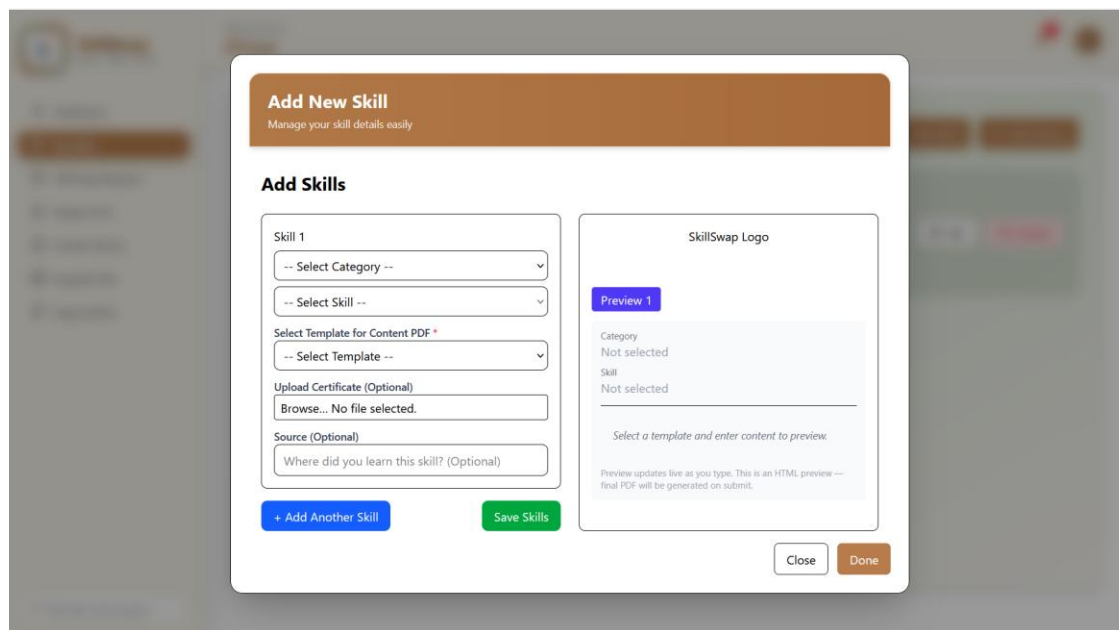


Figure 6.1.15: Add skill popup

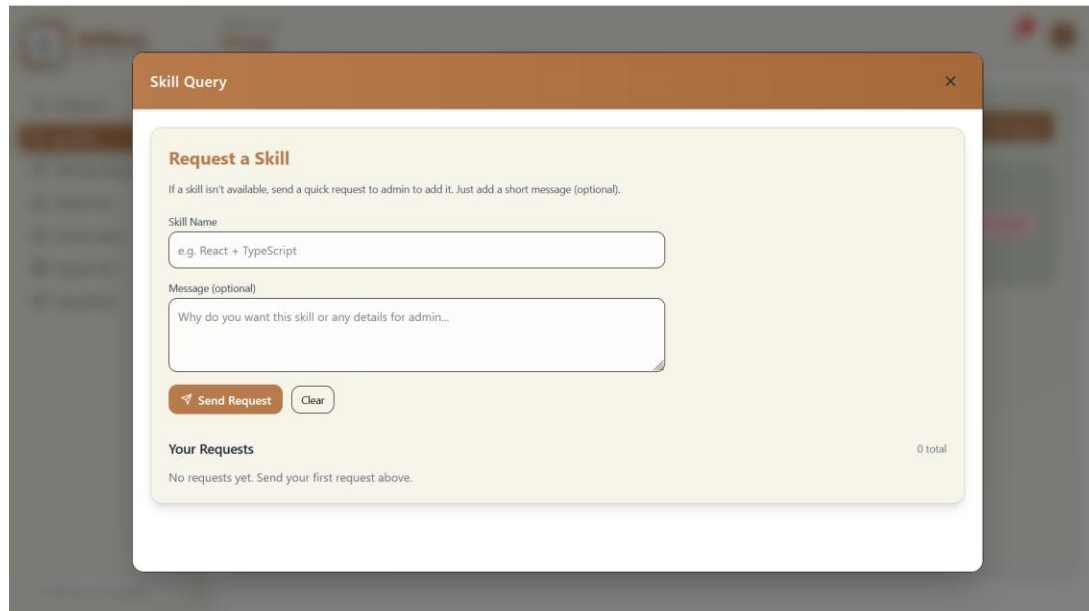


Figure 6.1.16: Request new skill popup

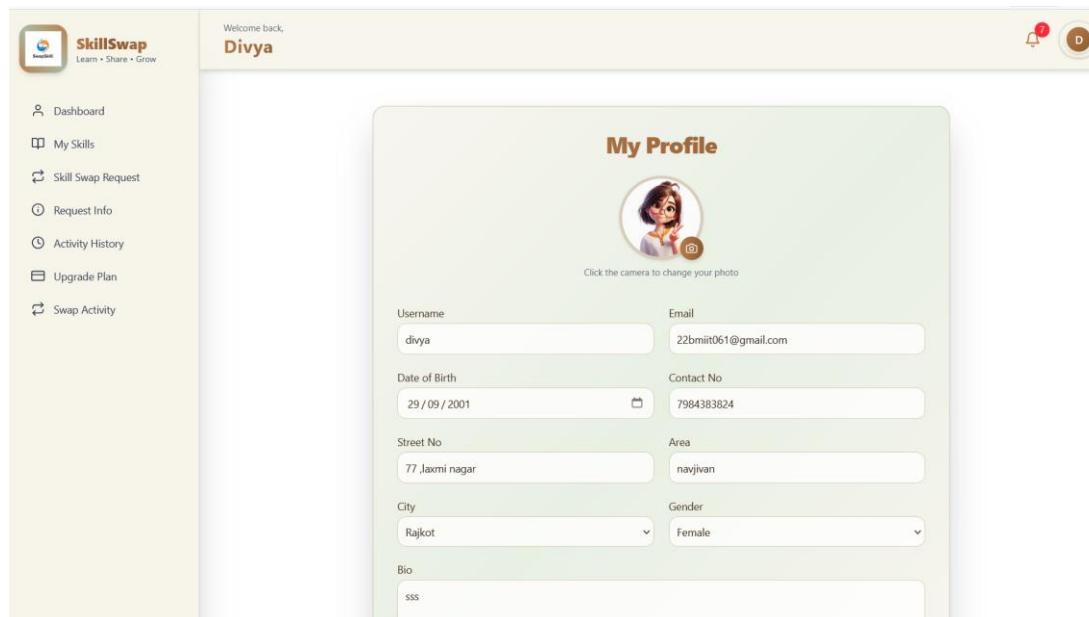


Figure 6.1.17: Update Profile Page

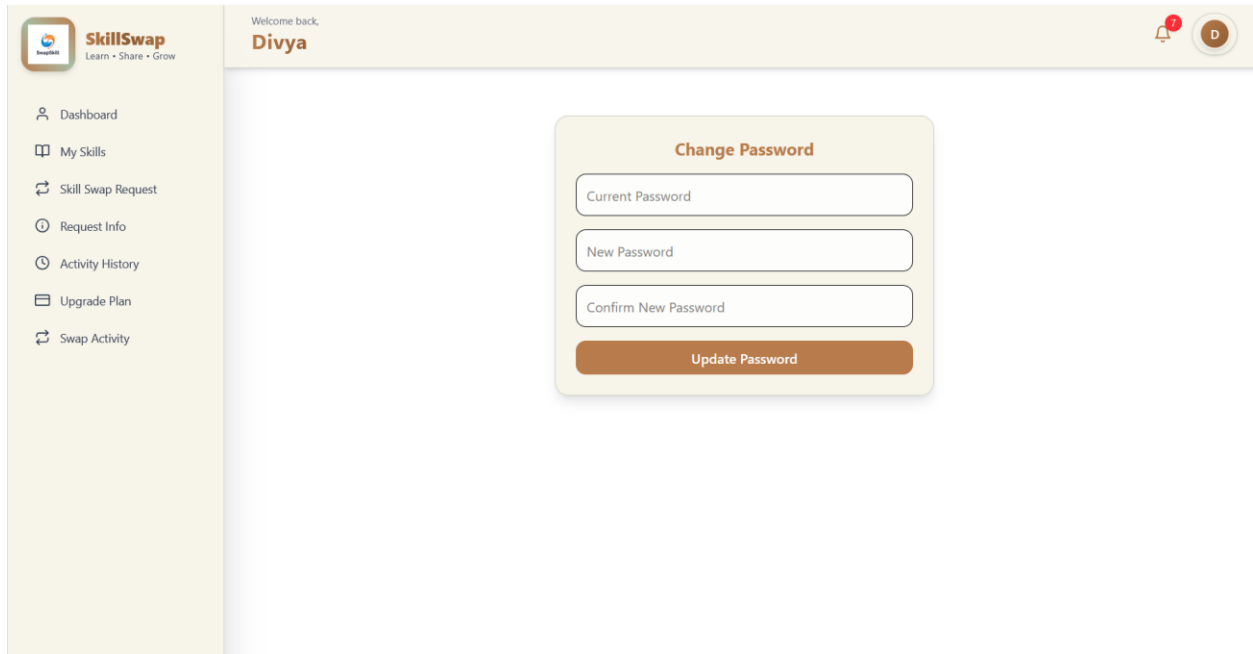


Figure 6.1.18: Change Password Page

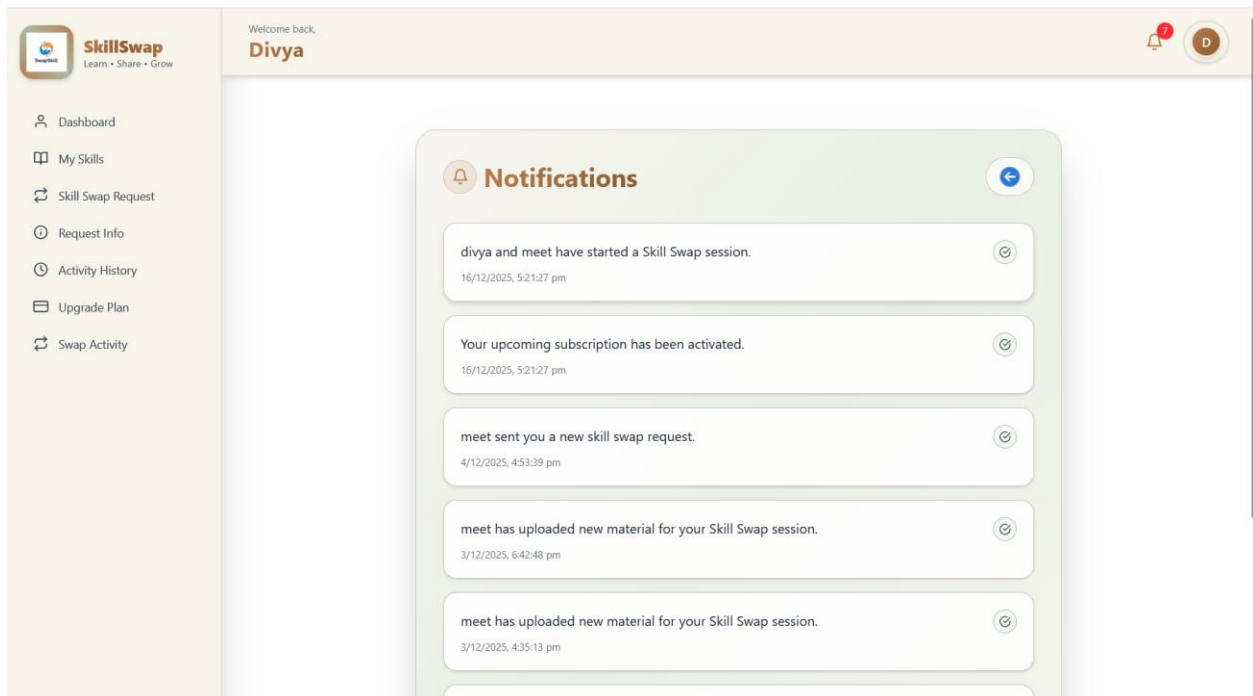


Figure 6.1.19: Notification Page

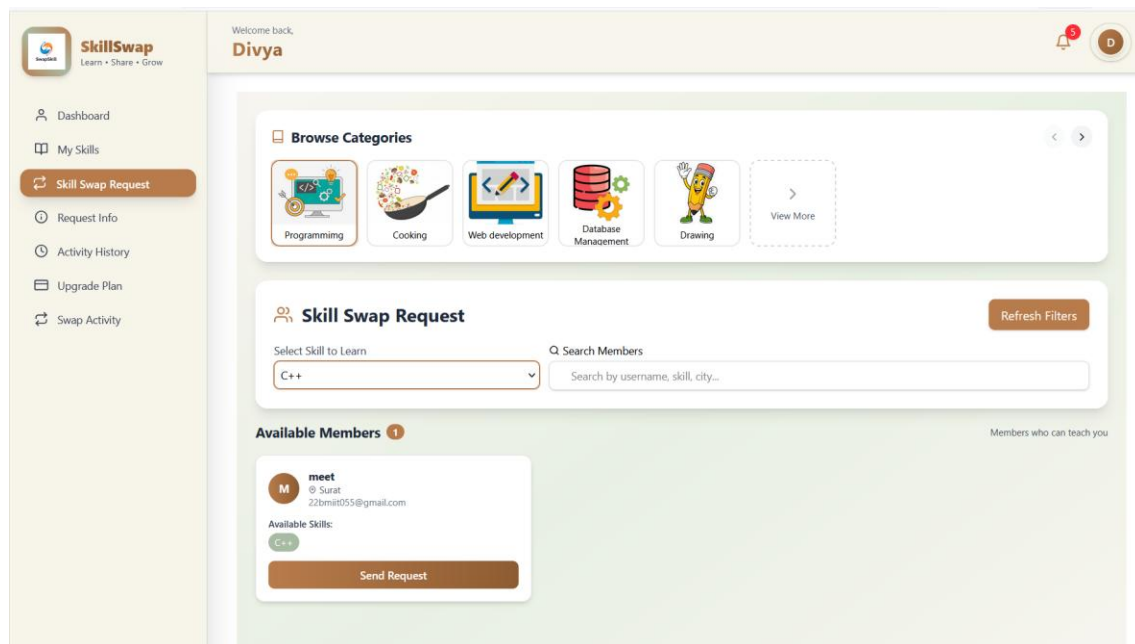


Figure 6.1.20: Send swap request to member page

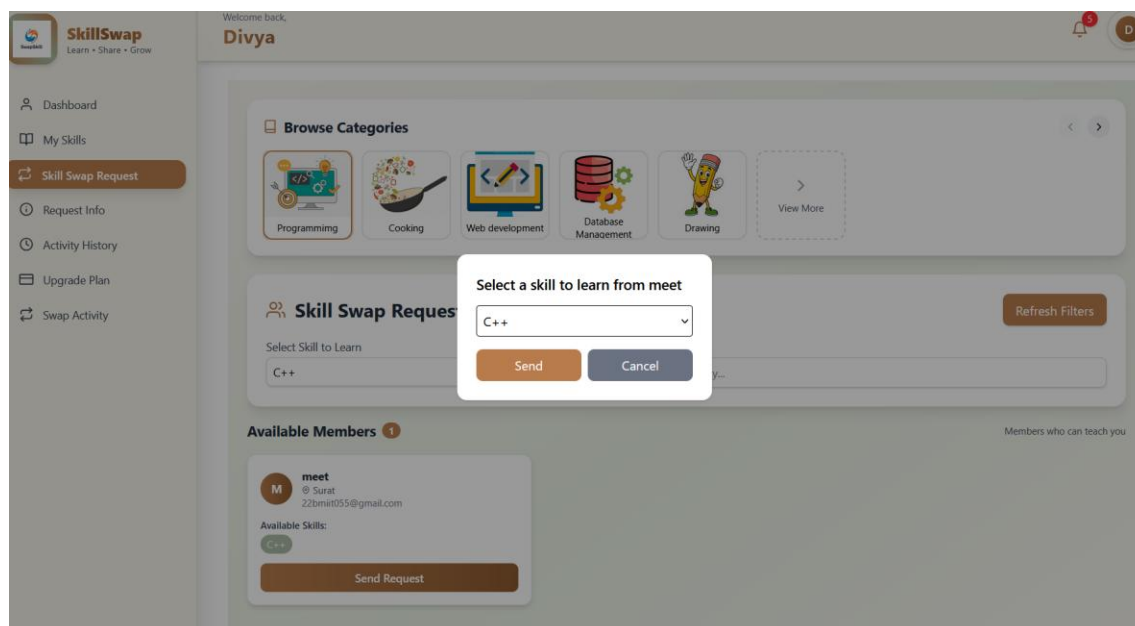


Figure 6.1.21: select skill popup while sending request

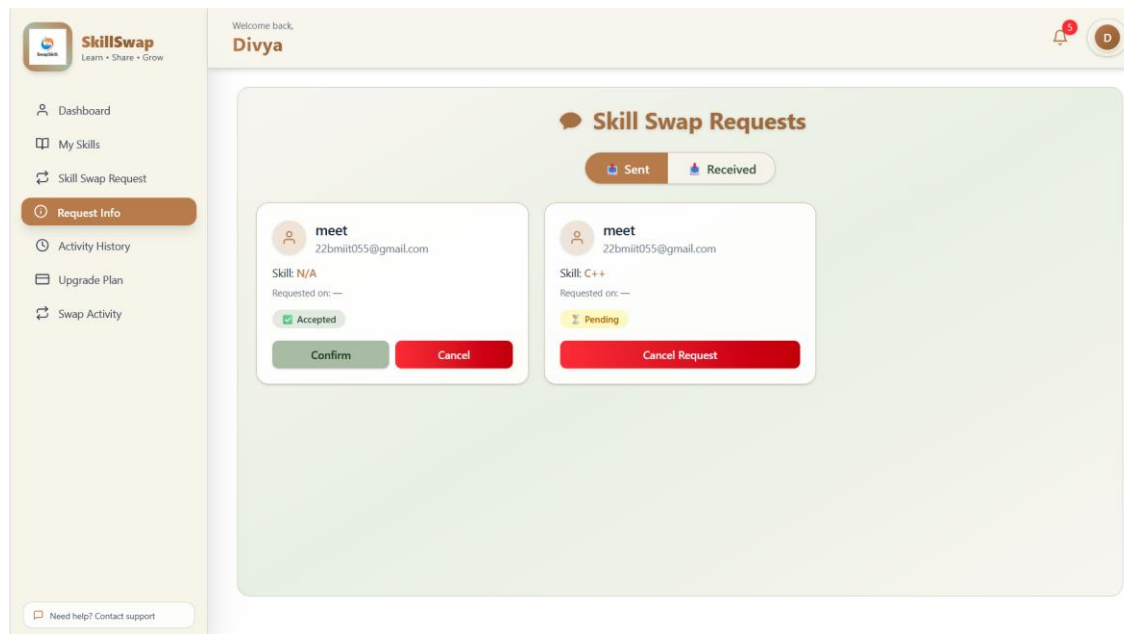


Figure 6.1.22: View sent/receive request page

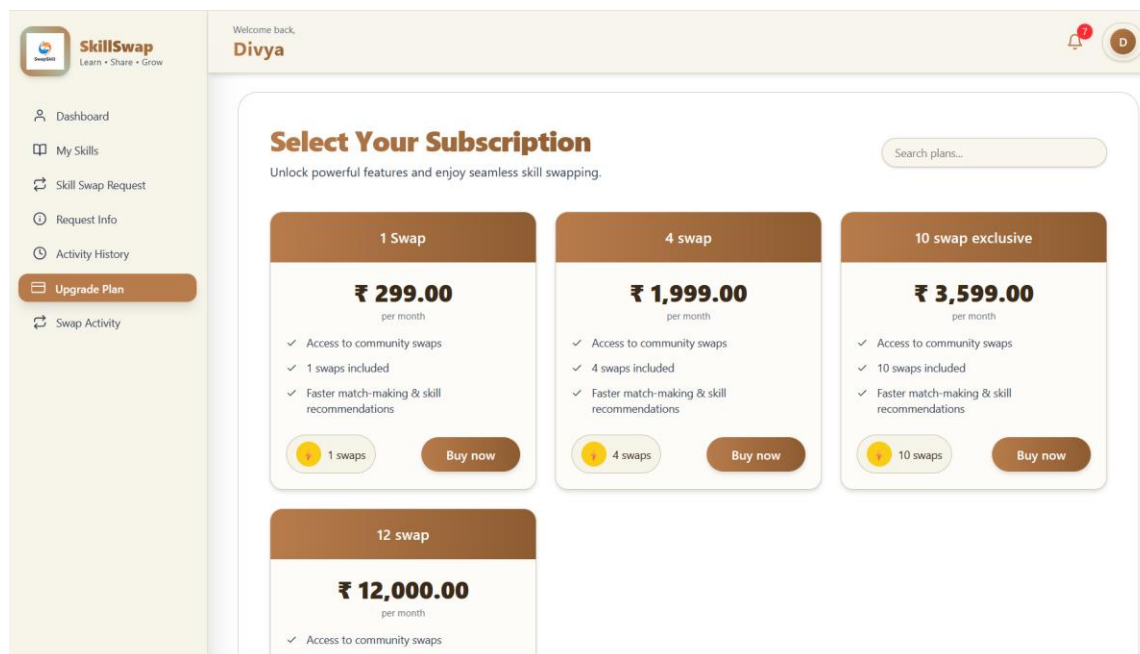


Figure 6.1.23: Purchase subscription page

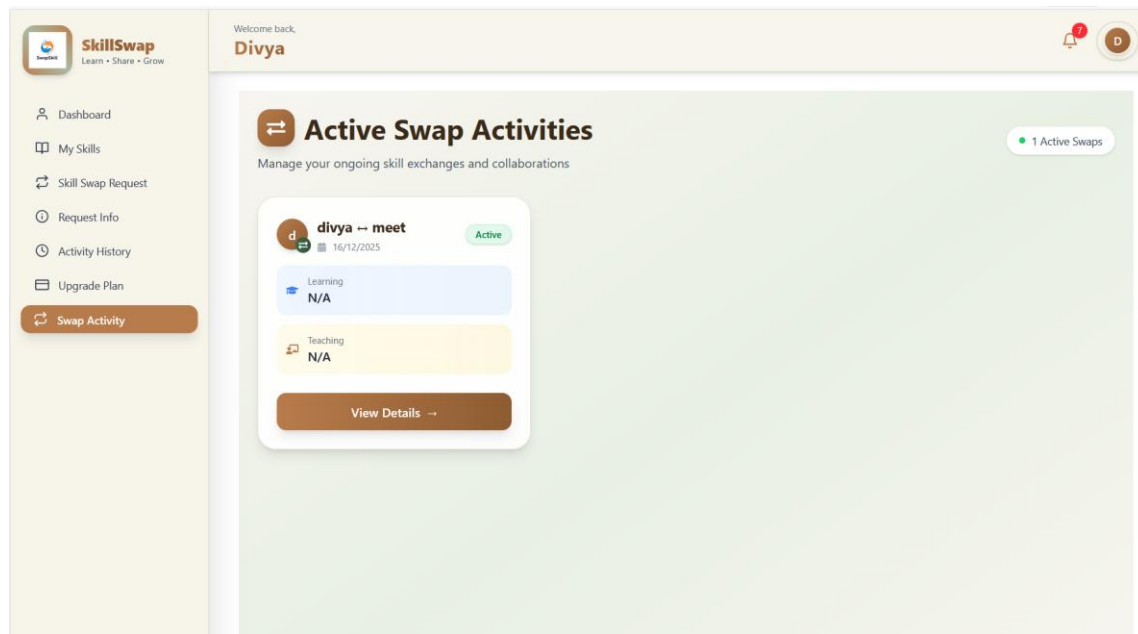


Figure 6.1.24: View Swap Activity page

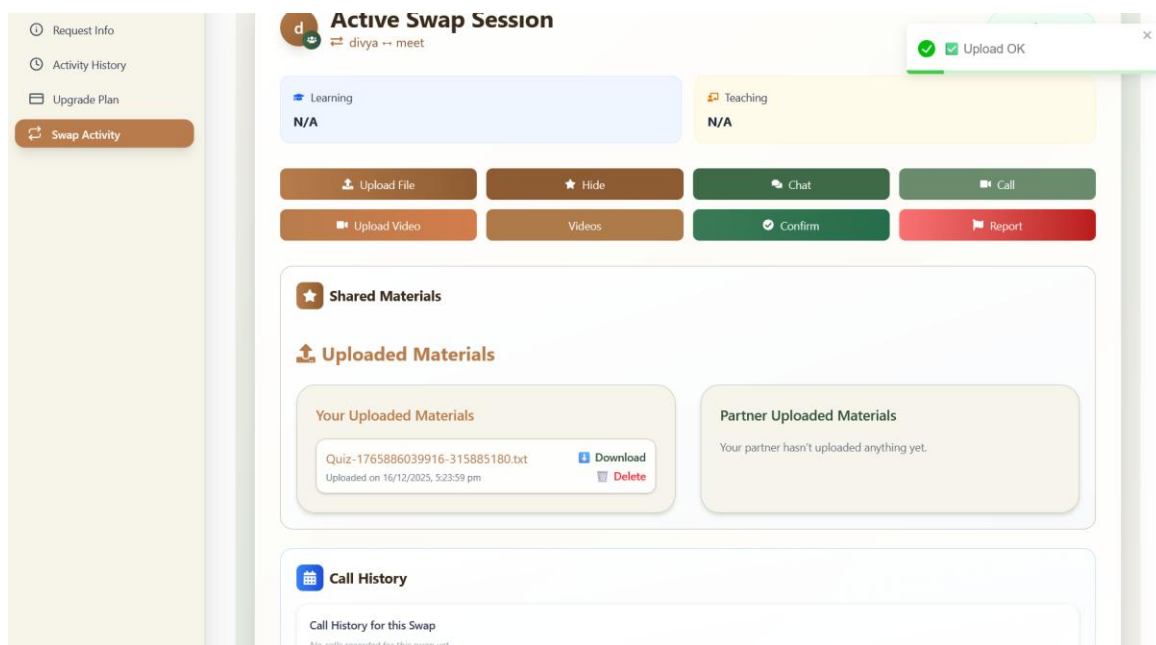


Figure 6.1.25: upload and show material feature page

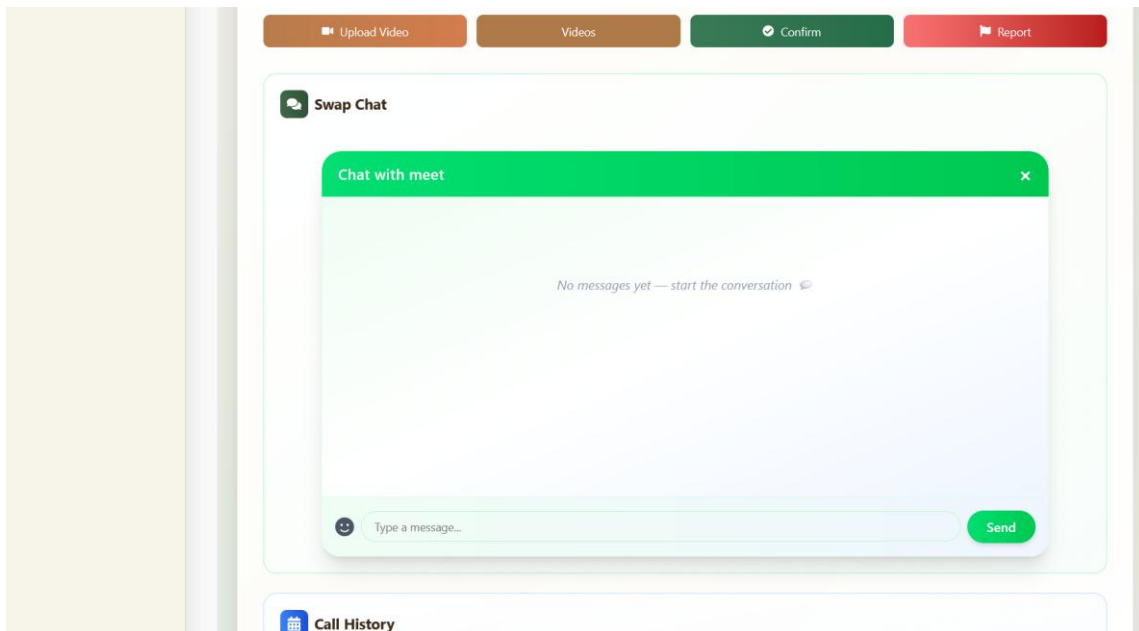


Figure 6.1.26: Chat feature page

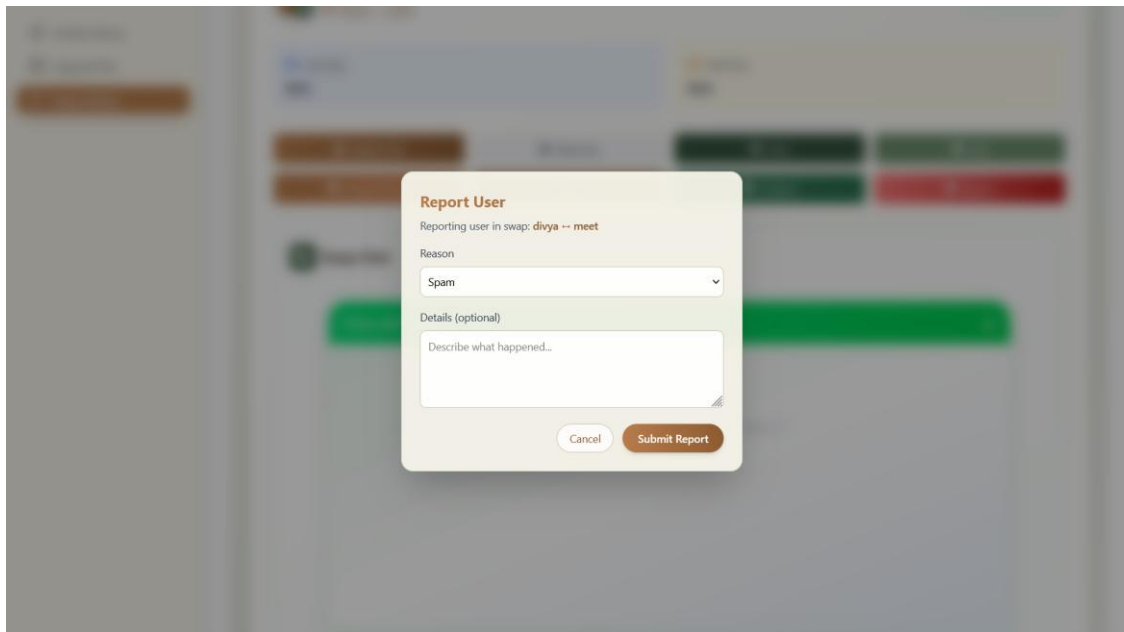


Figure 6.1.27: Report user(complaint) feature page

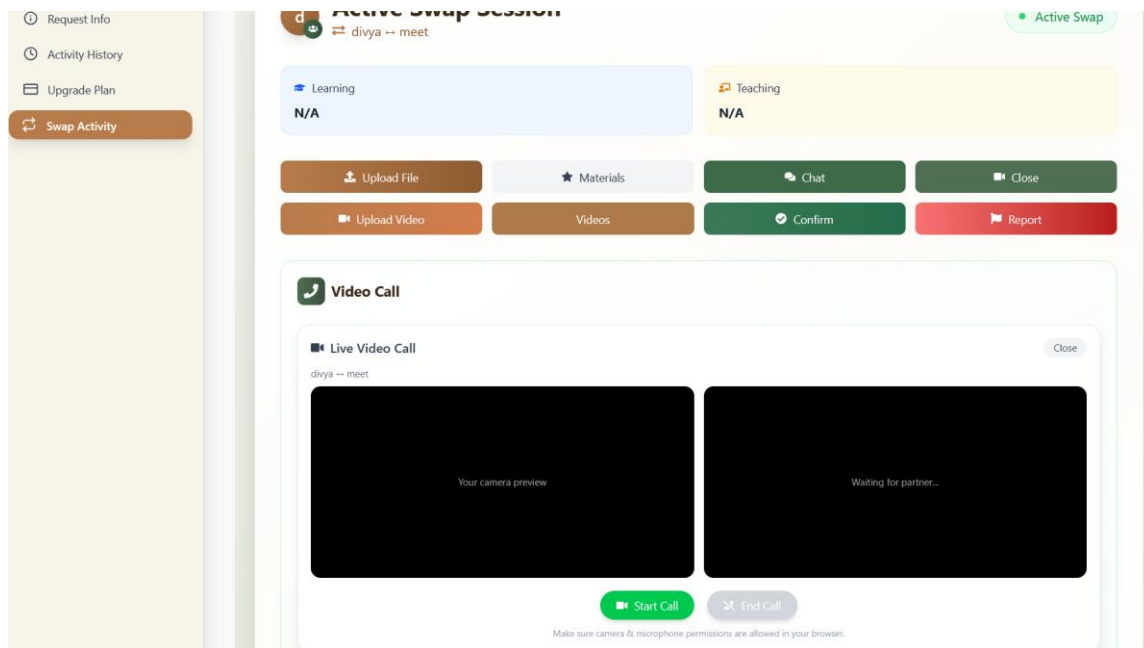


Figure 6.1.28: Video call feature page

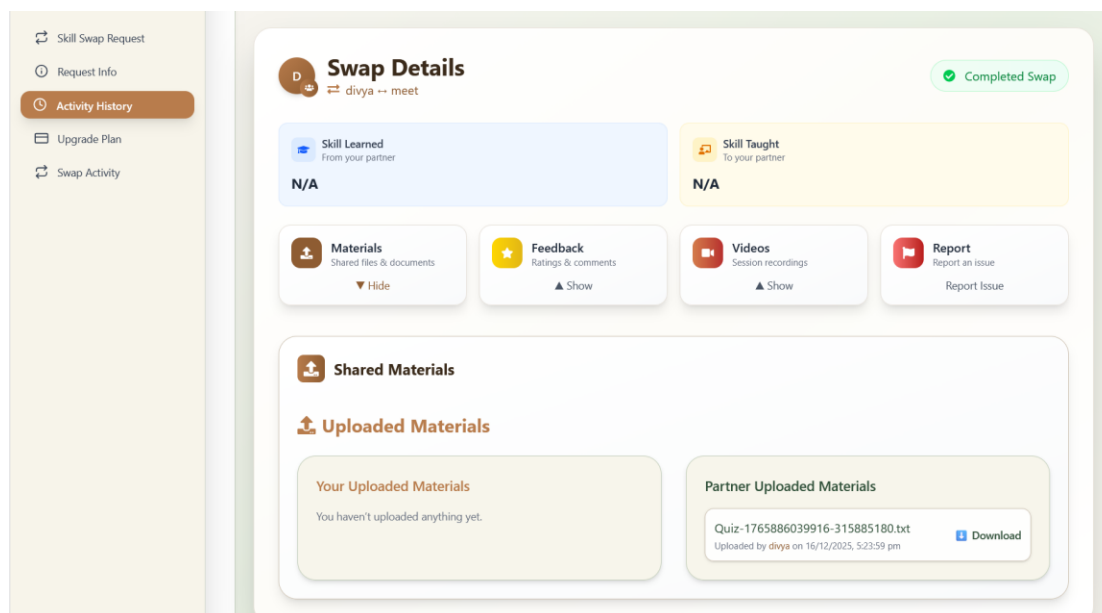


Figure 6.1.29: swap history page

★ Swap Feedback

Give Feedback

Rate your learning experience

✓ You already submitted feedback — You may update it.

Your Rating

★★★★☆ 4/5

Comments (optional)

"The teaching style was easy to understand and very practical."

63/500

Partner's Feedback

★★★★☆ 4/5

"It was a great skill swap session. I learned a lot and the explanation was very clear."

— meet

Cancel

Update Feedback

Figure 6.1.30: Feedback page

Chapter 7: Testing

7.1 Test cases

Test Case ID	Test Case	Inputs	Expected Output	Actual Output	Status
TC1	Verify Member Registration	Skills, Email, Profile Details	OTP sent to registered email	OTP sent to email	Pass
TC2	Verify OTP during Registration	Valid OTP	Registration successful and welcome email sent	Registration completed	Pass
TC3	Verify Login with valid credentials	Email, Password	User logged in successfully	Login successful	Pass
TC4	Verify Forgot Password	Registered Email	OTP sent to email	OTP received	Pass
TC5	Verify Profile Update	Updated profile details	Profile updated successfully	Profile updated	Pass
TC6	Verify Manage Skills, Cities, Subscription plan & Categories	Skills, Cities, Subscription plan & Categories details	Skills, Cities, Subscription plan & Categories added/updated/deactivate	Skills, Cities, Subscription plan & Categories managed	Pass
TC7	Verify Admin Blocks User	User ID	User status changed to blocked	User blocked	Pass
TC8	Verify View Subscription Payments	Admin login	Payment list displayed	Payments visible	Pass
TC9	Verify Send Skill Swap Request	Receiver ID, Skill	Request sent	Request sent	Pass
TC10	Verify Accept Skill Swap Request	Request ID, Skill	Request accepted	Accepted	Pass

TC11	Verify Reject Skill Swap Request	Request ID	Request rejected	Rejected	Pass
TC12	Verify Confirm Skill Swap	Both users confirm	Session starts	Session started	Pass
TC13	Verify Send Chat Message	Text message	Message delivered	Message sent	Pass
TC14	Verify Start Video Call	Call request	Video call started	Call connected	Pass
TC15	Verify Upload Session Material	File upload	Material shared with partner	File shared	Pass
TC16	Verify Submit Feedback & Rating	Rating, Comments	Feedback saved	Feedback submitted	Pass
TC17	Verify Report Inappropriate Behavior	Report details	Report sent to admin	Report submitted	Pass
TC18	Verify Purchase Swap Pack	Payment details	Payment successful	Payment done	Pass
TC19	View Skill Swap Activity History	Member login	Past swaps displayed	History shown	Pass

Chapter 8: Future Enhancement

Currently, the scope of this project is limited to users within a single state. In the future, the system can be expanded to support multiple states, allowing more users to connect and interact. This project can be improved in the future by adding new features and making the system handle more users easily. AI can be used to suggest suitable skill partners based on user interests, ratings, and previous skill swaps. The communication system can be improved by adding screen sharing, and instant notifications. The subscription system can also be improved by providing different plans, discount offers, and automatic renewal options.

Chapter 9: Conclusion

The Skill Swap platform provides a safe and easy way for users to share and learn skills without using money. Users can register, verify their accounts using OTP, search for skilled members, send requests, chat or video call, and share learning materials securely. Admin features like user management, complaint handling, skill control, and payment tracking help in maintaining proper system control. Feedback, ratings, and subscription features increase trust among users. Overall, the project fulfills all required functions and shows how modern web technologies can be used to solve real-life problems related to skill learning and sharing.

Chapter 10: Bibliography

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Node.js Official Documentation – <https://nodejs.org>

Express.js Documentation – <https://expressjs.com>

React.js Official Documentation – <https://react.dev>

Razorpay API Documentation – <https://razorpay.com/docs>

WebRTC Documentation for Video Calling – <https://webrtc.org>